

VLMs Use Feedforward or Feedback Visual Process

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Hypothesis

VLMs will perform well on degraded line drawings (local features survive) but fail on Mooney images (global completion required) — mirroring the feedforward-only architecture they implement.

The Phenomenon

Humans use top-down feedback: when local features are sparse, the brain predicts and completes global form from stored models.

Mooney images (pure B/W blobs) require gestalt completion —

Why Test VLMs?

VLMs process images in a single feedforward pass — no iterative top-down feedback at inference.

If recognition requires feedback, VLMs should fail on Mooneys even when they succeed on

Course Connection — Marr's Levels of Analysis

Computational level: Both humans and VLMs share the same goal — identify the object.

Algorithmic level: Humans use hierarchical inference with top-down feedback.
VLMs use feedforward feature matching in one forward pass.

Same input-output behavior does not imply the same algorithm.

Consistent with Marjieh et al. (2024): visual input adds no perceptual structure beyond what language priors already capture



er: Original (left) → Standard Mooney (center, both models wrong: 'wolf'/'pig') → Hard Mooney (right, both wrong: 'cow'/'cow')

SPEAKER SCRIPT — Slide 1: Motivation

~45 seconds · Do not read verbatim — use as a guide

We're testing whether vision-language models process images the way humans do — specifically whether they use top-down feedback or pure feedforward matching.

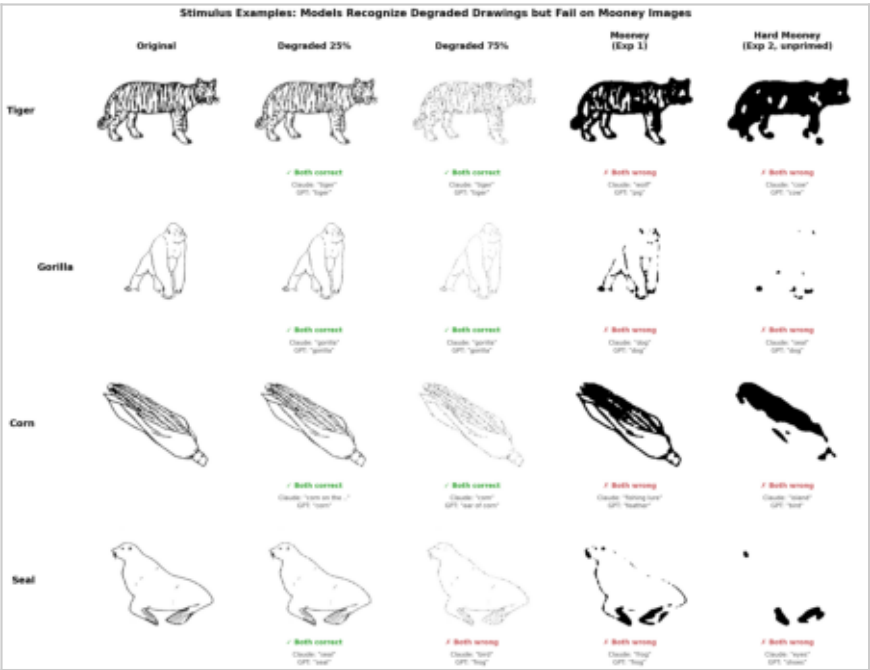
The Mooney image is our key tool. It's a black-and-white blob that humans can't identify at first — until their visual system completes the global form. It requires feedback from stored object knowledge.

VLMs don't have that feedback loop — they process images in one forward pass. So our hypothesis was: VLMs will do fine on degraded drawings, where local features survive, but fail on Mooneys, where global completion is required.

This connects directly to Marr's levels — same computational goal, potentially very different algorithm.

✂ **Key point: Point at the tiger example — 'Both models got this completely wrong on Mooney'**

Methods



Design

Stimuli: 260 Snodgrass & Vanderwart (1980) canonical B&W line drawings
Models: claude-sonnet-4-5 (Anthropic) · gpt-5.4 (OpenAI) via Brown CCV LiteLLM
Trials: 4,160 total · Checkpointed after every trial for crash recovery

Exp 1 — 6 conditions: Degraded 25 / 50 / 75 / 90% · Standard Mooney ($\sigma=2.0$) · Inverted Mooney
Exp 2 — 2 conditions: Hard Mooney unprimed ($\sigma=4.0$) · Hard Mooney primed (degraded_25 first)

Confound Controls

Contamination control: Inverted Mooney (B/W flipped) — if models memorized training images, accuracy should drop steeply. Small drop → not contamination.

Language prior control: Identical prompt across all conditions and both models — prevents autoregressive artifacts from inflating scores.

Reproducibility: Fixed random seed (42) for all degradation; full protocol in code.

Human baselines: Snodgrass & Corwin (1988) · Moore & Cavanagh (1998)

SPEAKER SCRIPT — Slide 2: Methods

~45 seconds · Do not read verbatim — use as a guide

We used 260 Snodgrass and Vanderwart line drawings — the canonical stimulus set from the human agnosia literature, so we can compare directly to published norms.

We ran two experiments. Experiment 1 tested six conditions: four levels of pixel degradation, a standard Mooney image, and an inverted Mooney as a contamination control.

Experiment 2 added harder Mooneys — stronger blur — with and without a priming image shown first.

Both models got the exact same prompt, which controls for language priors. We also checked that the small inverted Mooney drop rules out training data memorization.

4,160 total trials, saved after every single one.

✂ **Key point: Emphasize: same prompt for all conditions — this is the key confound control**

Exp 1 Mooney: Claude 82% · GPT 80% · Human 34% | Exp 2 Unprimed: Claude 27% · GPT 22% · Human ~20% | Priming boost: Claude +53% · GPT +67% · Human ~+45%



Figure 1. Experiment 1: Accuracy across all conditions. Human baselines from Snodgrass & Corwin (1988).

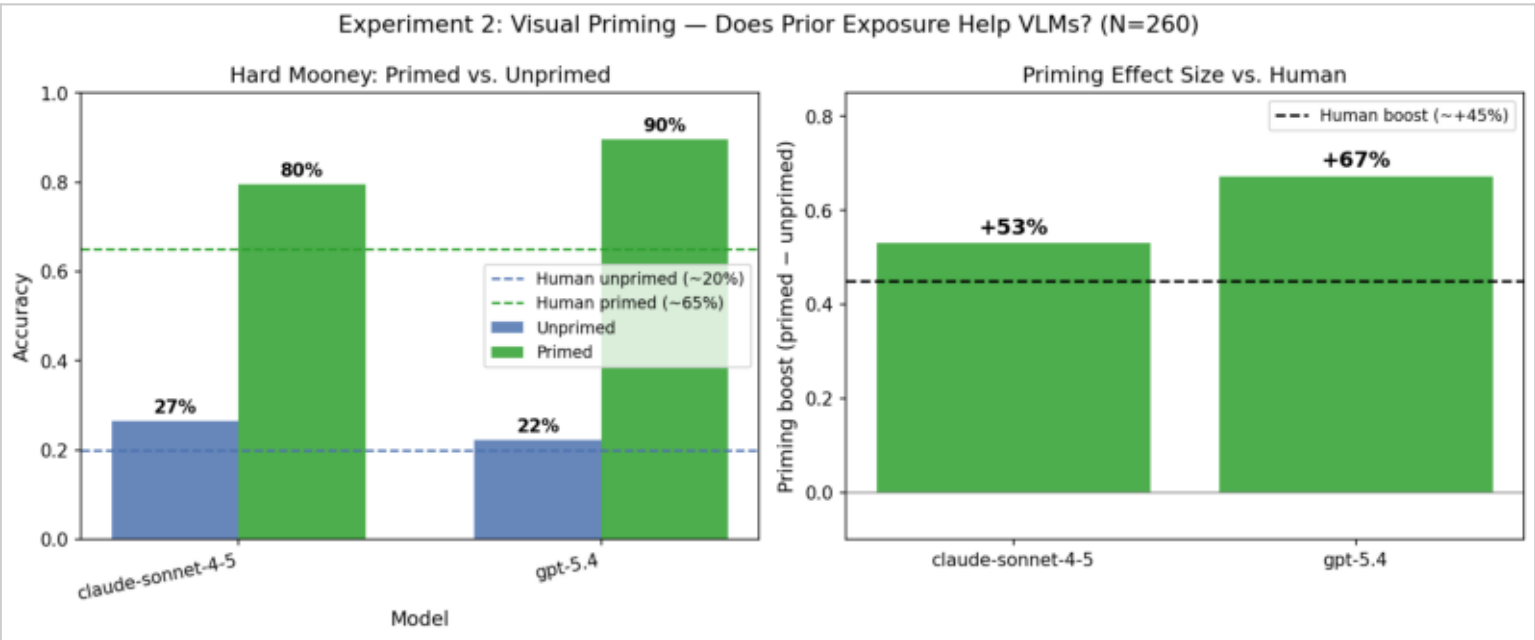


Figure 2. Experiment 2: Hard Mooney primed vs. unprimed. Human priming boost ~+45% (Moore & Cavanagh, 1998).

SPEAKER SCRIPT — Slide 3: Results

~45 seconds · Do not read verbatim — use as a guide

Three main findings.

First — both models beat humans on degraded drawings at every level. That's consistent with feedforward feature matching.

Second — on standard Mooney images, both models scored around 80%, versus 34% for humans. That's the opposite of what we predicted. But it turns out many S&V objects have such distinctive silhouettes that they survive thresholding — banana, tiger, airplane. No global completion needed.

Third — Experiment 2. When we used stronger blur, unprimed accuracy dropped to human levels: 27% and 22%. Then we showed the degraded image first — priming — and accuracy jumped over 80% for both models. That's a bigger boost than humans get.

✂ **Key point: Point at Fig 1 first, then Fig 2. Stress: unprimed drops to human level = feedforward shortcut gone**

Open Questions & Conclusions

Finding 1 — Feedforward shortcut

VLMs exceed humans on degraded drawings because feedforward feature matching doesn't require global form. At 90% degradation, GPT still scores 80% vs. ~15% human — a striking dissociation.

Finding 2 — Mooney difficulty is stimulus-dependent

Standard Mooney: ~80% vs. 34% human. S&V silhouettes are too distinctive — no gestalt completion required. Hard Mooney ($\sigma=4.0$) eliminates this: unprimed accuracy drops to human levels.

Finding 3 — Priming $\neq$ perceptual memory

Both models show +53–67% priming boost — larger than humans. But both images are in one forward pass. This is in-context feature co-matching, not top-down memory. Same behavior, different mechanism.

Marr's Levels — Prediction vs. Understanding

Same computational goal (name the object). Different algorithm (feedback vs. feedforward). Aggregate accuracy can look similar while mechanisms are fundamentally different. Consistent with Marjeh et al. (2024): VLMs predict labels without perceptual understanding.

Limitations

- Priming within one API call — not true memory
- S&V silhouettes reduce Mooney difficulty
- Human baselines use different degradation

What We'd Do Next

- Separate-call priming (true memory test)
- Face Mooneys (non-diagnostic silhouettes)
- Explain Claude vs. GPT gap at 90% degradation

SPEAKER SCRIPT — Slide 4: Conclusions

~45 seconds · Do not read verbatim — use as a guide

Three takeaways.

One: VLMs use feedforward feature matching. They beat humans on degraded drawings because surviving pixels are enough — they don't need the complete picture.

Two: Standard Mooneys were too easy because S&V silhouettes are distinctive. Hard Mooneys fixed that — unprimed accuracy dropped to human levels.

Three: The priming boost looks like a human effect, but it's not the same thing. Humans prime across separate viewings — perceptual memory. VLMs just compare two images that happen to be in the same forward pass.

This is Marr's levels in action: same output, different algorithm. VLMs predict the label without building a perceptual representation of the object.

⚡ **Key point: End with the Marr point — this is the course connection. Be ready for: 'is priming really different?'**

Appendix — Supplementary Analysis

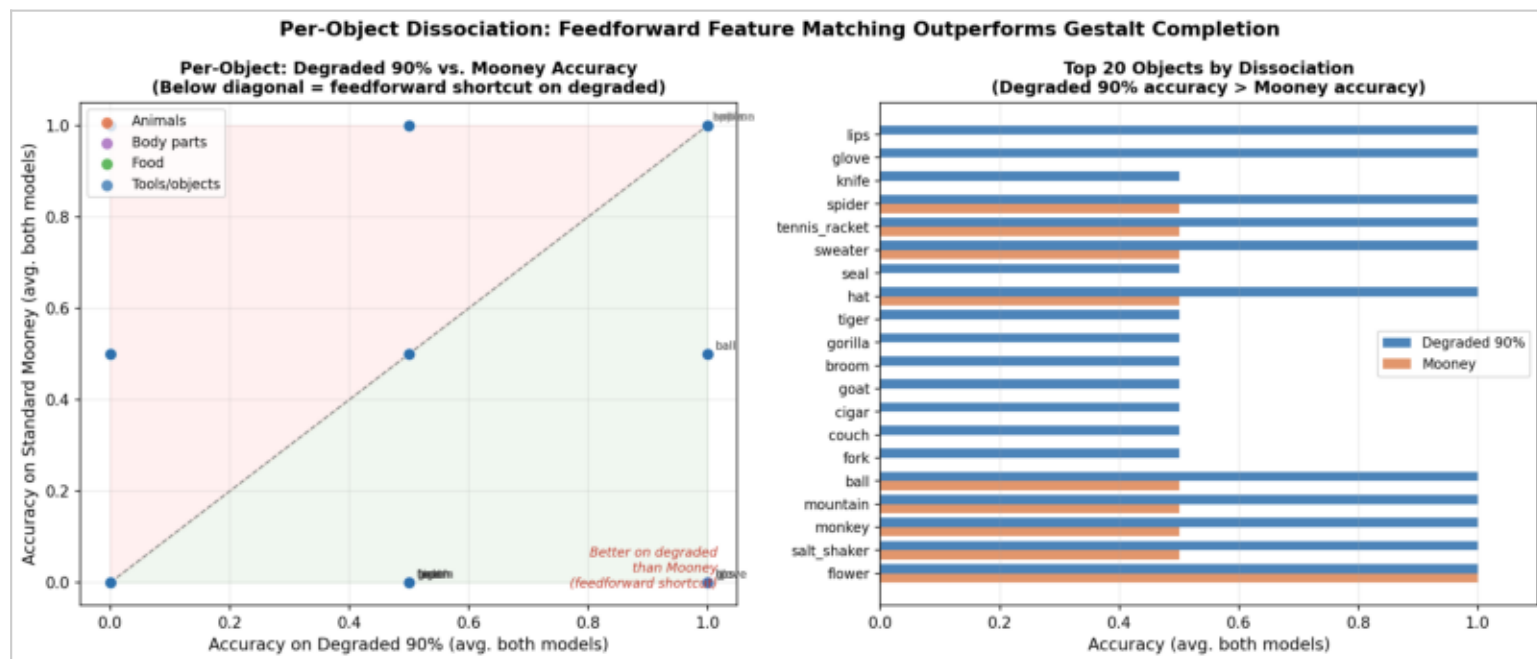
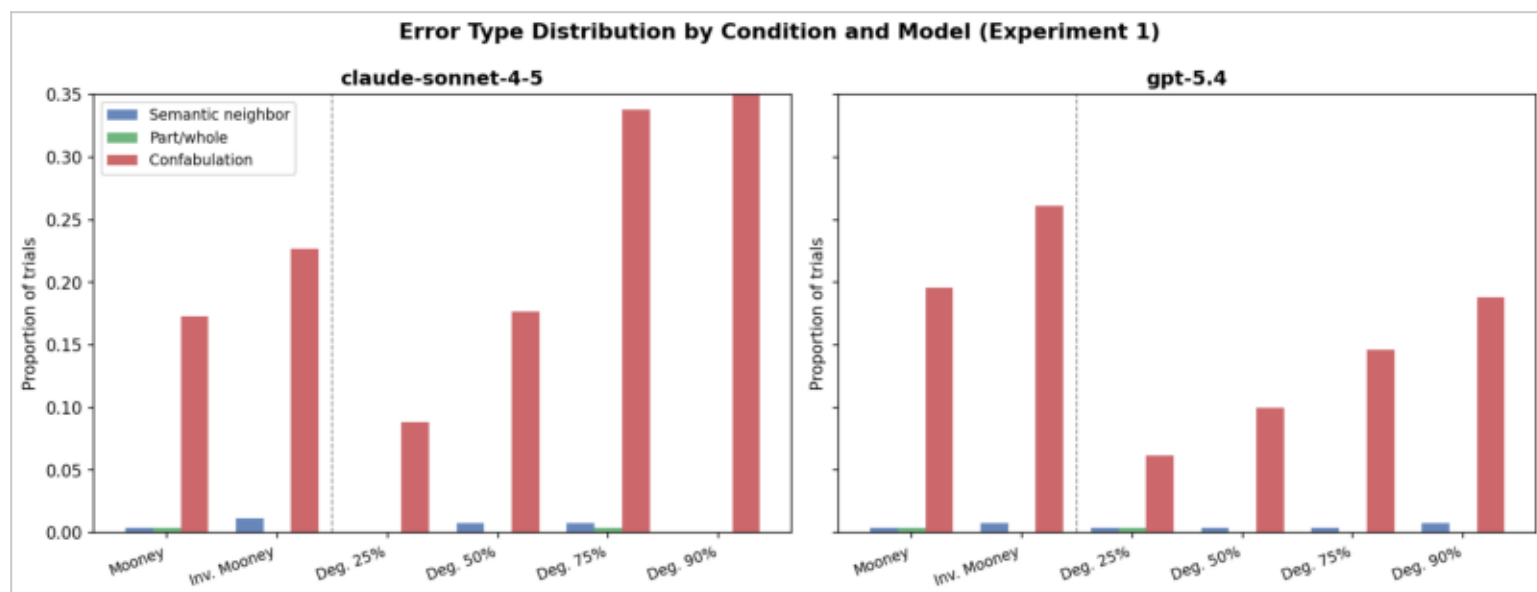


Figure 4. Per-object dissociation: degraded 90% vs. Mooney accuracy (avg. both models, N=260 objects).



pe distribution by condition and model. Confabulation dominates — models produce wrong objects, not semantic

SPEAKER SCRIPT — Slide A: Appendix

~45 seconds · Do not read verbatim — use as a guide

This is the supplementary slide — don't present it unless someone asks during Q&A.

Figure 4 shows the dissociation at the per-object level. Each dot is one of the 260 S&V objects. Points below the diagonal did better on degraded 90% than Mooney — that's the feedforward shortcut in action at item level. The right panel shows the top 20 most dissociated objects.

Figure 3 shows error types. Nearly all errors are confabulation — the model names a completely unrelated object, not a semantically related one. This suggests the failure is perceptual, not semantic — the model can't form a stable representation to even attempt category-level retrieval.

✂ **Key point: Only reference if asked: 'Can you show more detail on individual objects?' or 'What kind of errors?'**